

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 2.4: SURFACE AREA AND VOLUME OF SOLIDS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have spent this sub-strand investigating the puzzle from the Kisumu water tank factory: two tanks that each hold 5,000 litres of water, yet the frustum tank uses less metal sheet than the cylinder-hemisphere tank. Use your notes, your formulae, your Comparison Table and your cumulative model to answer each question below. Write in full sentences, show your working where calculations are needed, and use the correct mathematical words (surface area, volume, slant height, net, composite solid, frustum). Use the data in the table in Section 1 whenever a question refers to it. Metal sheet costs KSh 850 per square metre.

Section 1: Reading the evidence from the Kisumu factory

The table below shows the results you calculated for the two tanks made at the Kisumu factory for households in Siaya County.

Tank design	Volume of water held	Total surface area	Metal sheet cost at KSh 850 per m ²
Cylinder with hemispherical lid	approx. 5,000 litres	approx. 16.09 m ²	approx. KSh 13,677
Frustum	approx. 5,000 litres	approx. 7.03 m ²	approx. KSh 5,975

- (a) Which tank needs less metal sheet? By roughly how many square metres, and by roughly how much money, does it differ from the other tank?
- (b) Explain in your own words why the students at the factory were puzzled. Use the words volume and surface area in your answer.

Write your answer here:

Section 2: Why surface area and volume are independent

- (a) What does 'surface area' of a solid tell a manufacturer, and what does 'volume' tell a manufacturer? Give one example of each from the water tank factory.
- (b) Use your work on the two tanks to explain what it means to say that surface area and volume are independent properties of a solid.
- (c) A cone and a cylinder have the same base radius and the same vertical height. Compare their volumes and compare their total surface areas. What does this comparison show about how shape affects each quantity?

Write your answer here:

Section 3: Explaining the result using formulae and nets

- (a) The total surface area of the frustum tank is found using $SA = \pi(R^2 + r^2) + \pi(R + r)l$. Name the part of the tank that each of the three terms πR^2 , πr^2 and $\pi(R + r)l$ represents.
- (b) When you found the surface area of the cylinder-hemisphere tank, you worked with a composite solid. Describe the three steps of the strategy you used and explain what happens at the joint between the cylinder and the hemisphere.
- (c) Using your answers above and the table in Section 1, explain why the frustum tank has the smaller surface area for the same volume.

Write your answer here:

Section 4: Engineer's recommendation for Nyakach Secondary School

You are a junior engineer at a Kenyan water tank manufacturer. Nyakach Secondary School in Siaya County needs a tank that holds 5,000 litres. Metal sheet costs KSh 850 per square metre.

- (a) Using the table in Section 1, show how the metal sheet cost of the frustum tank is calculated from its surface area.
- (b) Which of the two designs in the table would you recommend to the school? Justify your choice with at least two pieces of mathematical evidence.
- (c) Name one other thing, apart from metal sheet cost, that an engineer or the school might want to consider before the final choice is made.

Write your answer here:

Section 5: Using this mathematics in Kenya

- (a)** Describe one real situation in Kenya, other than the Kisumu water tanks, where someone needs to know the surface area of a solid, and one where someone needs to know its volume. Name the solid or solids involved.
- (b)** Choose one career (for example architecture, agriculture, manufacturing or water engineering) and explain how the idea that shape affects surface area and volume independently helps a person in that career to make good decisions.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses the table values and the Kisumu tank data accurately and refers to them in every answer, including correct differences and costs.	Uses most of the table values correctly, with minor slips in differences or costs.	Uses little of the data or quotes values incorrectly.
Understanding that surface area and volume are independent	Clearly explains with the two tanks and the cone and cylinder comparison that equal volume can come with different surface areas, and that shape affects each quantity differently.	States that surface area and volume are independent and supports it with one example.	Confuses surface area with volume or says a larger volume always means a larger surface area.
Mathematical reasoning with formulae and strategies	Correctly identifies each term in the frustum formula, describes the three-step composite solid strategy including hidden faces, and links these to the result.	Identifies most terms and steps, with small gaps in explaining hidden faces or slant height.	Names formulae or steps incompletely or incorrectly.
Justified recommendation and Kenyan applications	Gives a clear recommendation supported by at least two pieces of evidence, and gives accurate, relevant Kenyan examples and a career link.	Gives a recommendation with some evidence and at least one suitable Kenyan example.	Gives a recommendation without evidence or gives examples that are unclear or unrelated.

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